

Institute of Actuaries of India

Subject ST6 – Finance and Investment B

March 2017 Examination

INDICATIVE SOLUTION

Solution 1:

i)

An asset-backed security (ABS) is a security whose income payments and hence value are derived from and collateralized (or “backed”) by a specified pool of underlying assets. The pool of assets is typically a group of small and illiquid assets which are unable to be sold individually. Pooling the assets into financial instruments allows them to be sold to general investors, a process called securitization, and allows the risk of investing in the underlying assets to be diversified because each security will represent a fraction of the total value of the diverse pool of underlying assets. The pools of underlying assets can include common payments from credit cards, auto loans, and mortgage loans, to esoteric cash flows from aircraft leases, royalty payments and movie revenues.

(4)

ii)

Price	2000
Gry	
	Property
1	100
2	100
3	100
4	2431.0125

There would be lease cashflows and payment after

(4)

iii)

	Property	Bond	Corporate	Bond	Property-void	Corporate-After default	Total-Assets
1	100	100	100	100	96	98	294
2	100	100	100	100	96	98	294
3	100	100	100	100	96	98	294
4	2431.0125	1000	1000	1000	2431.013	980	4411.013

(4)

iv) Net cashflows

	Total Assets	Total Liability	Net
1	294	0	294
2	294	1500	-1206
3	294	0	294
4	4411.013	2700	1711.013

(2)

v)

Asset valuation would require use of asset reinvestment rate and overdraft rates

	Total Assets	Total Liability	Net	Cummulative	Interest charges	Cashflows
1	294	0	294	294	15	309
2	294	1500	-1206	-897	-90	-1296
3	294	0	294	-708	-71	223
4	4411.013	2700	1711.013	1,022	51	1762

Discounting cashflows at 10% for the value of the asset
£581

(6)

vi)

- Regulatory risk
- Default risk
- Operational risk
- Change of liability cashflows
- Change in asset cashflows
- Expenses

(3)

[23 Marks]

Solution 2:

i)

A swaption is a combination of the following two financial instruments: Interest Rate Swap and Option. Swaption9 (Swap Option) reserves the right for its holder to purchase a swap at a prescribed time and interest rate in the future (European Option). The holder of such a call option has the right, but not the obligation to pay fixed in exchange for variable interest rate. Therefore, this option is also known as "Payer Swaption". The holder of the equivalent put option has the right, but not the obligation to receive interest at a fixed rate (Receiver Swaption) and pay variable.

The company should purchase receiver swap with strike of +50bps given the risk appetite

(3)

ii)

Strike Swap Rate (s_k)	0.0550	Time	curve	Factor
Forward Swap Rate Continuous Compounding	0.0600	5.0	0.06	0.000000
Forward Swap Rate m -Annual Compounding (s_0)	0.0609	5.5	0.06	0.718924
Volatility of Forward Swap Rate (σ)	0.20	6.0	0.06	0.697676
Notional Principal (L)	100	6.5	0.06	0.677057
Number of Payments per year (m)	2	7.0	0.06	0.657047
Sum of Discount Factors (A)	2.0036	7.5	0.06	0.637628
Option Maturity (T)	5	8.0	0.06	0.618783
Payer Swaption (P) or Receiver Swaption [®]	R			

Swaption Price	1.51
d1	0.45
D2	0.00
Swaption Price	1.514867

(5)

iii)

Selling a payer swaption at interest rate +50bps would pass on any profit company makes with increase in interest rate over 50bps to buyer of the payer swaption. This transaction would results in initial premium which would compensate the cost of the receiver swaption.

(8)

iv)

Strike Swap Rate (s_k)	0.0650	Time	curve	Factor
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Option Maturity (T)	5	8.0	0.06	0.618783
Payer Swaption (P) or Receiver Swaption (R)	P			
Swaption Price	1.85			
d1	0.08			
D2	-0.37			
Swaption Price	1.845071			

Difference is 0.330204 CR which is profit from setting up the collar

The breach of risk appetite can be corrected by purchasing a payer swaption with strike of +150bps on current yield. The payer option sold at 50bps would result in loss till +150bps after that the new payer swaption would compensate for the loss.

			LIBOR yield curve	Discount Factor
Strike Swap Rate (s_K)	0.0750	Time		
Forward Swap Rate Continuous Compounding	0.0600	5.0	0.06	0.000000
Forward Swap Rate m -Annual Compounding (s_0)	0.0609	5.5	0.06	0.718924
Volatility of Forward Swap Rate (σ)	0.20	6.0	0.06	0.697676
Notional Principal (L)	100	6.5	0.06	0.677057
Number of Payments per year (m)	2	7.0	0.06	0.657047
Sum of Discount Factors (A)	2.0036	7.5	0.06	0.637628
Option Maturity (T)	5	8.0	0.06	0.618783
Payer Swaption (P) or Receiver Swaption (R)	P			
Swaption Price	1.25			
d1	-0.24			
D2	-0.69			
Swaption Price	1.248255			

Total cost for the set up is 0.91805

(7)

[23 Marks]

Solution 3:

$$\begin{aligned}
 d(s_1 s_2) &= \{s_1 s_2 \mu_1 + s_1 s_2 \mu_2 + \rho_{12} \sigma_1 \sigma_2 s_1 s_2\} dt + \sigma_1 s_1 s_2 dz_1 + \sigma_2 s_1 s_2 dz_2 \\
 d(s_1 s_2) &= s_1 s_2 [\mu_1 + \mu_2 + \rho_{12} \sigma_1 \sigma_2] dt + s_1 s_2 (\sigma_1 dz_1 + \sigma_2 dz_2) \\
 d(s_1 s_2) &= s_1 s_2 [\mu_1 + \mu_2 + \rho_{12} \sigma_1 \sigma_2] dt + s_1 s_2 (\sigma_3 dz_3)
 \end{aligned}$$

[4 Marks]

Solution 4:**i)****a)** Price of a 5-year 10% annual coupon bond at current rates:

$$P = 10 \sum_{i=1}^5 d_i + 100d_5 = 10(0.95238 + 0.90488 + 0.8577 + 0.81492 + 0.77796) + 100(0.77796) \\ = 120.875$$

where d_i are the values of the zero coupon bonds of length i (per nominal of 1 unit).

(2)

b) The value of a 5-year annual fixed-floating swap with 10% fixed coupon is the difference between a 10% 5-year bond and an FRN. But the latter is valued at 100.

Hence swap value = 120.875 – 100 = 20.875.

[This is a receiver swap – the payer swap is of opposite sign.]

(2)**ii)**

a) *Modified Duration*, $M = -\frac{1}{P} \frac{\partial P}{\partial r}$

where r is the level of rates (or yields).

$$\text{Absolute Yield Sensitivity} = \frac{P_r - P_{r+\Delta r}}{\Delta r}$$

So, Absolute Yield Sensitivity of the Bond = (Modified Duration * Price of the Bond), if Δr is small and the entire yield curve shifts in parallel.

(2)

b) First we need to calculate the absolute yield sensitivity of the bond and swap. Using the same technique as in (i), we get:

Bond price (shifted) = 10 (4.296) + 100 (0.77428) = 120.388.

Hence Absolute Yield Sensitivity of the bond = [(120.875 – 120.388) / 0.10] = 4.87.

Similarly, the swap values (shifted) = 120.388 – 100 = 20.388.

Hence Absolute Yield Sensitivity of the swap = [(20.875 – 20.388) / 0.10] = 4.87.

(3)**iii)**

a) The cash flows of the reverse floater are:

Coupons:	Years 1 to 5: $10 - f_i$	$i = 1$ to 5	(A)
Redemption:	+ Year 5: 100		(B)

But the coupons on an annual fixed – floating swap are the same as in (A) and the 5-year zero coupon bond pays 100 at the end of year 5 as in (B).

Hence, the decomposition as required.

(3)

b) Price and absolute yield sensitivity are additive.

So the price of the reverse floater = swap + 5 year zero = 20.875 + 77.796 = 98.671.

Also, the absolute yield sensitivities = swap + 5 year zero = 4.873 + {(77.796 – 77.428) / 0.10}
= 4.873 + 3.680 = 8.553

So, Modified Duration of reverse floater = [(8.553 * 100) / 98.671] = 8.668, which is roughly twice the Modified Duration of the par yield bond (4.299).

(4)

iv) There are two possible alternative answers, both equally valid.

Bond world

In a bond world, the portfolio is risk managed by ensuring that modified durations match. The total duration is the sum of the individual bonds, weighted by nominal amount. Longer bonds will have greater convexity. Swaps are treated as a long (or short) fixed bond position and an equal short (or long) floating bond position, and their durations obtained separately. The reverse floater is decomposed into a swap and zero coupon bond using the analysis in (iii) above.

OR

Swap world

In a swap world, the risk management is performed by sensitivity analysis to the underlying yield curve. The bonds are mapped to the relevant yield curve points by taking their price sensitivity to each of the components of the curve in turn, i.e. 1 year zero, 2 year zero, etc. Swaps are decomposed into fixed and floating legs and the inverse floater is decomposed as in (iii).

(3)

[19 Marks]

Solution 5:

- i) Fund = 275 million, Index = 1,100. Therefore, Fund = INR 250,000 times the index.
Fund falls 5% iff index falls 5% (To 1045 from the current level of 1100).
So puts of \$250,000 times the index with strike 1,045 are required
Apply Black Scholes with:
 $S_0 = 1,100$; $K = 1,045$; $T = 1$; $r = 5\%$ (risk free rate); $\sigma = 25\%$; $y = 3\%$ (dividend yield)

Further,

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - y + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} \text{ and } d_2 = d_1 - \sigma\sqrt{T}$$

This implies, $d_1 = 0.41017$ and $d_2 = 0.16017$. Thus, $N(-d_1) = 0.34084$ and $N(-d_2) = 0.43657$.

Value of one put = $K e^{-rT} N(-d_2) - S_0 e^{-yT} N(-d_1) = 69.9236$

Thus, the total cost of insurance = INR(250,000 * 69.9236) = INR 17,481,000

(4)

- ii) From put-call parity (or by re-arranging the Black Scholes equations for valuing puts and calls) we have

$$V_{Put} = V_{Call} - S_0 e^{-yT} + K e^{-rT}$$

This shows that a put option can be created by shorting e^{-qT} X the index, buying a call option (of same term and strike as the put) and investing the remainder in the risk free asset.

For fund manager under question, this involves:

- Selling INR 275m e^{-yT} = INR 266,872,500 of stock
- Investing in INR 250,000 X the S&P-500 index of calls with strike 1,045 and a term of 1 year.
- Investing the remainder in 1-year risk free zero coupon bonds yielding 5%.

(4)

- iii) The delta of one put option is

$$e^{-yT} (N(d_1) - 1) = e^{-0.03 \times 1} (0.65916 - 1) = -0.33077$$

Hence, this implies that 33.077% of the portfolio (i.e INR 90.962m) needs to be sold (shorted) and the proceeds invested in risk free assets.

(3)

- iv) The delta of a 9 – month index futures contract is

$$e^{(r-y)T} = e^{(0.05 - 0.03)0.75} = 1.0151.$$

The short position required is (90.962 million / 1100) = 82,693 times the index.

Hence a short position involving (82,693 / (1.0151 * 250)) = 325.85 (rounded to 326) futures contracts is needed.

(3)

[14 Marks]

Solution 6:

- i) a)

Using Black-Scholes to price the Call option:

$$d1 = (\ln(500/600) + (5\% + 0.5 \times 22.5\% \times 22.5\%)) / 22.5\% = -0.4756$$

$$d2 = d1 - 22.5\% = -0.7006$$

$$\text{So, } C = 500 \times N(-0.4756) - 600 \times e^{(-0.05 \times 1)} \times N(-0.7006) = 20.60$$

Similarly, using the Black-Scholes to price the Put option:

$$d1 = (\ln(500/400) + (5\% + 0.5 \times 22.5\% \times 22.5\%)) / 22.5\% = 1.3265$$

$$d2 = d1 - 22.5\% = 1.1015$$

$$\text{Thus, } P = 400 \times e^{(-0.05 \times 1)} \times N(-1.1015) - 500 \times N(-1.3265) = 5.33$$

(5)

b)

Let the four risk-neutral probabilities be p_1, p_2, p_3, p_4 with p_1 at the bottom of the tree as shown in the diagram.

An asset that pays X_i in i th scenario is worth $[X_1p_1 + X_2p_2 + X_3p_3 + X_4p_4] e^{(-0.05 * 1)}$

Bonds will be priced correctly because:

$$[e^{(0.05*1)}p_1 + e^{(0.05*1)}p_2 + e^{(0.05*1)}p_3 + e^{(0.05*1)}p_4] e^{(-0.05*1)} = 1$$

For the Call to be priced properly: $[0p_1 + 0p_2 + 0p_3 + 100p_4] e^{(-0.05*1)} = 20.60$

So, $p_4 = 0.217$

For the Put to be priced properly: $[100p_1 + 0p_2 + 0p_3 + 0p_4] e^{(-0.05*1)} = 5.33$

So $p_1 = 0.056$

Note that p 's add up to 1, so $p_3 = 1 - 0.056 - 0.217 - p_2 = 0.727 - p_2$

For equities to be priced properly:

$$[300*0.056 + 450p_2 + 550(0.727 - p_2) + 700*0.217] e^{(-0.05*1)} = 500$$

Thus, $p_2 = 0.428$ and $p_3 = 0.299$.

(5)

ii) These answers are in reference to the calculations in part (i).

Deriving the p 's involves solving four simultaneous equations as before, but with different values for the Call and the Put.

The Call option has lower volatility so will be lower in price and since the Call goes down in value, p_4 will be lower.

The Put option has higher volatility so will be higher in price and since the Put has become more expensive, p_1 will be higher.

To quantify impact on p_2 and p_3 , we note that:

- $p_2 + p_3$ is constrained by the four probabilities summing to 1
- the expected one-period equity price in the risk neutral measure will be unchanged, i.e. $[300*p_1 + 450p_2 + 550p_3 + 700*p_4] e^{(-0.05*1)} = 500$

Now, if we need to leave the expected value unchanged after p_1 rises and p_4 falls, we need to increase weighting on scenario 3 at the expense of scenario 2.

Hence p_2 will be lower and p_3 will be higher.

(4)

iii)

- For the low end of the equity prices: The distribution with the volatility skew will have a fatter tail as compared to the distribution without the volatility skew
- For the high end of the equity prices: The distribution with the volatility skew will have a thinner tail as compared to the distribution without the volatility skew
- For the mean distribution: The mean of the distribution with the volatility skew will be shifted to the right (towards higher equity prices) as compared to the one without the volatility skew

(3)

[17 Marks]
